

RESEARCH ARTICLE

Terrestrial Insect Communities and the Restoration of an Industrially Perturbed Landscape: Assessing Success and Surrogacy

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Abstract

Restoration has been acknowledged to be a major pathway by which humans can repair damaged ecosystems. Most of the work to date on terrestrial systems has focused on restoring plant communities, with an assumption that the conditions that lead to more diverse plant communities will also lead to a restoration of insect communities. This, however, has not been proven to be true. Here, we examine the recovery of terrestrial insect communities in naturally recovering and restored sites in response to severe historic

pollution in the region of Sudbury, Ontario. We find that the insect communities have not recovered to the same degree as the plant communities. Furthermore, we present directions on how cross-taxa surrogacy can be useful to further aid the use of insects as indicators of restoration success.

Key words: ants, Carabidae, copper–nickel smelter, diversity, Formicidae, ground beetles, mining pollution, restoration success, richness, terrestrial insects.

Introduction

Industrial barrens from the long-term smelting of copper and nickel are found in many regions around the world (e.g. Canada, Finland, Russia) and span thousands of hectares (Kozlov & Zvereva 2007). The resulting derelict land has relatively low levels of diversity and abundance in many taxa, including plants (Hutchinson & Whitby 1977; Winterhalder 1996), insects (Kozlov et al. 2005), and vertebrates (Kataev et al. 1994). Despite reduced pollution levels (or the removal of the pollution source entirely), recovery of these ecosystems may be slow due to legacy effects such as lingering concentrations of heavy metals and hydrogen ions (acidity) in the soil (Gundermann & Hutchinson 1995) and the persistence of thin soil layers susceptible to erosion (Hutchinson & Whitby 1977; Winterhalder 1996; Anand et al. 2003).

Initial restoration efforts in these regions have focused on liming, seeding, and tree planting and have been successful in terms of increasing the plant diversity of these ecosystems (Lautenbach et al. 1995; Winterhalder 1996; Kiikkilä 2003 and references therein; Rayfield et al. 2005). Recent studies on other regions have shown that the arthropod community diversity has lagged behind restoration of plant community diversity

(Longcore 2003; Majer et al. 2007). As a result, restored sites have been called “depauperate imitations,” as they superficially resemble the undisturbed sites in some ways but do not have the same arthropod diversity or composition (Longcore 2003). Some studies have suggested that insect communities may be correlated to plant species richness and plant complexity (Brose 2003; Pearce et al. 2003). However, restoration of the diversity of the plant community does not always imply restoration of the full range of fauna and biological processes. An increase in plant diversity does not necessarily result in an increase in insect diversity due to interactions with competitors, predators, and parasitoids (Bommarco & Banks 2003; Perner et al. 2003) as well as effects on changing plant composition in restored sites. Anand et al. (2005) found that although vascular plants diversity correlated with the bryophyte community diversity in unassisted sites of the Sudbury region, restored sites showed no relationship between the two taxonomic groups, possibly due to the introduction of non-native plant species.

Restoring insect communities is important due to the roles that various insect groups play in ecosystem structure and function. The presence of certain insect species can reveal particular aspects of ecosystem dynamics. Ants (Hymenoptera: Formicidae) can significantly affect soil nutrients, temperature and aeration, and seed dispersal, which in turn may alter the vegetation of the area (MacMahon et al. 2000; Majer et al. 2007). In areas with thin or nutrient-poor soils, ants can be essential for soil composition and revegetation after disturbance and have even been considered more important than earthworms for a number of soil processes (Lyford 1963).

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In the temperate North American forests, ants are predators of forest defoliators like the spruce budworm (Sanders & Pang 1992) and are dispersers of forest herb seeds like *Viola* sp. and *Trillium* sp. (Beattie & Culver 1981). Carabid beetles (Coleoptera: Carabidae) are primarily predators (Ball & Bousquet 2001), although other feeding guilds (phytophages and omnivores) within the family are often overlooked (Harvey et al. 2008). Certain carnivorous species may indicate the presence of particular prey items or act as surrogates for the biodiversity of other beetle families. Both carabids and ants have been used as bioindicators of stress and disturbances such as clearcutting, fire, and mining practices (Majer 1983; Jennings et al. 1986; Heliölä et al. 2001; Rainio & Niemelä 2003; Teixeira et al. 2005; da Silva et al. 2008).

Insects are considered one of the most feasible taxa to work with for a range of ecological studies (Lawes et al. 2005), and ants and carabid beetles, in particular, are further considered the most useful study organisms within terrestrial insects (Beaudry et al. 1997; Majer et al. 2004). This is due to factors such as their abundance, diversity of guilds or species, and quick response to environmental change (Majer 1983; Beaudry et al. 1997; Rainio & Niemelä 2003). Despite their importance, biodiversity assessment and monitoring of insects are often not feasible for short-term studies due to the intensive taxonomic identification and research costs. Surrogacy may therefore be explored. The two common approaches of surrogacy for insects are cross-taxon surrogacy (using one taxonomic group to represent others) and using higher taxa (family, genus, etc.) as surrogates for lower taxa (species). Studies have suggested that using higher taxa is a better form of surrogacy for invertebrates (more congruency or correlation) than cross-taxon, allowing for more rapid evaluation by parataxonomists (Lovell et al. 2007). However, if cross-taxon surrogacy is used, it should only be used to represent taxa within the same trophic guild and a multi-taxa approach should be used to cover a broad range of guilds (Lovell et al. 2007; Majer et al. 2007).

In this article, we compare the plant and insect community of sites recovering naturally (unassisted) from historic pollution to those of sites under a restoration (assisted) regime near Sudbury, Canada. Our objective is to understand how restoration efforts may be affecting the terrestrial insect community and at what taxonomic level these effects can be observed. By comparing the results of the various taxa, we address the potential of the surrogacy approach.

Methods

Site Description

The focus of this article is on the region of Sudbury, Ontario, Canada, which was once considered one of the most ecologically disturbed regions in Canada due to mining and smelting practices of the late 1800s and early 1900s that decimated practically all woody plants in the region and resulted in the contamination and erosion of the soil (Winterhalder 1995). The restoration techniques used in the Sudbury area include liming, seeding, and planting trees and have been successful in the

recovery of plants in many areas (Winterhalder 1996; Rayfield et al. 2005). The ecological impact of such pollution has been documented for various taxa, including plants (Bačkor & Fahselt 2004; Anand et al. 2005; Rayfield et al. 2005), aquatic invertebrates (Yan et al. 2004), fish (Pyle et al. 2005), soil microorganisms (Anand et al. 2003), and mites (St John et al. 2002). A few studies that have focused on terrestrial insects in this area aimed to examine the effects of fragmentation on herbivore outbreaks (Roland 1993) and the accumulation of metals in the insect herbivores (Bagatto & Shorthouse 1996). To our best knowledge, no studies have used ants or carabids as the primary study organisms in assessing restoration success in this region.

Four study sites were selected from the Sudbury region (46°27.5'N to 46°28.6'N and 80°51.0'W to 80°53.1'W) in Ontario, Canada. The area is part of the Boreal Shield of Canada in The Great Lakes–St. Lawrence forest region surrounding Sudbury, Ontario. The Great Lakes–St. Lawrence forest is a climatically transitional forest between Boreal forest to the north and the temperate deciduous forest further south, often dominated by white pine (*Pinus strobus*), white birch (*Betula papyrifera*) and red pine (*Pinus resinosa*) (Scott 1995). With a temperate climate, the summer temperature in the Sudbury region reaches over 30°C and winter temperatures fall below –30°C. The soils are mainly podzol or gleysol. However, years of acidification and erosion have left many areas with a very thin acidic soil layer.

The sites are located near a smelter complex that was decommissioned in the 1970s. There are two localities, Coniston and Daisy Lake, with two sites within each locality, an unassisted site and an assisted site (Fig. 1). Both unassisted sites are naturally recovering and are characterized by a very thin or absent soil layer and a pH ranging from 3.7 to 4.6. The Coniston unassisted site is located approximately 1.7 km from the smelter and the vegetation is composed mainly of grasses, mosses, and white birch where it is not bare ground or rock. The Daisy Lake unassisted site is approximately 4.2 km from the smelter complex and has distinctly more soil, plant species, and ground cover than Coniston but remains relatively open with few trees.

The assisted sites have had various restoration techniques applied to them. The Coniston assisted site is approximately 0.7 km from the smelter complex and has undergone over 23 years of rehabilitation through liming, fertilizing, and seeding. Jack pine, red pine, white pine, and red oak trees were also planted at this site within 30 m of our transect. The Daisy Lake assisted site is 3.6 km from the smelter complex and has had over 12 years of rehabilitation from aerial liming, seeding, and planting of Jack pine.

We use previously manipulated sites on the landscape which we feel have allowed a sufficient time for insect communities to recover. The selected assisted sites have obviously not received exactly the same restoration regimes. However, our objective here is not to compare different restoration regimes, but to compare assisted with unassisted sites. These sites represent the best opportunities to study assisted along with

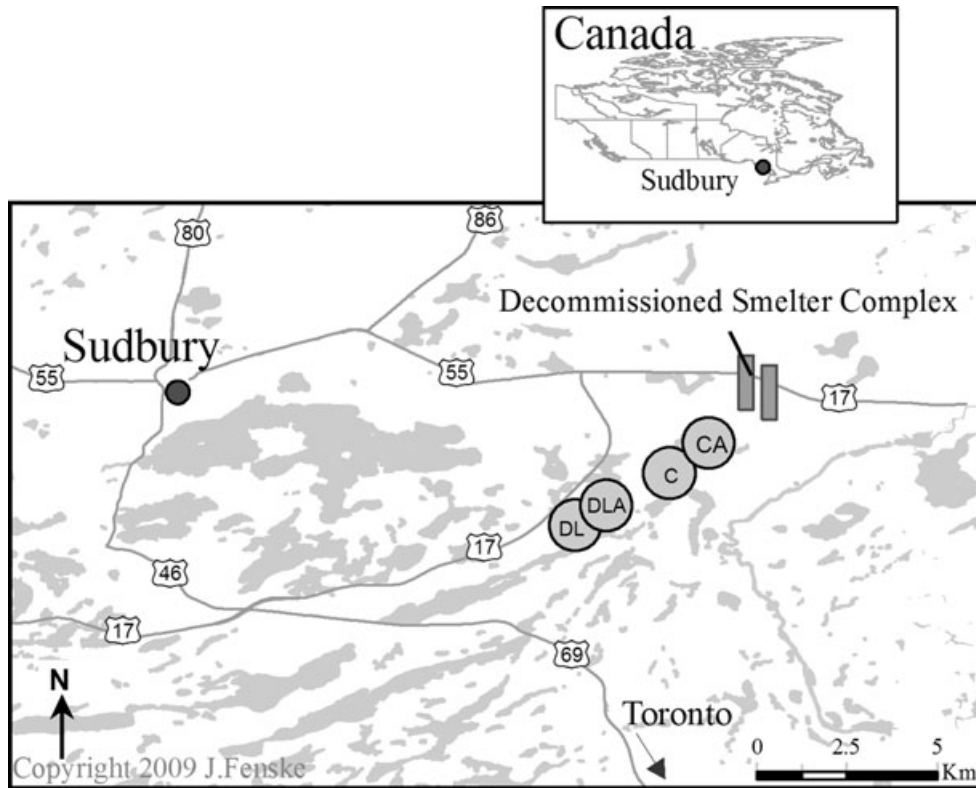


Figure 1. Map showing the locations of the four study sites and the decommissioned smelting complex near Sudbury, Ontario, Canada. Sites include Coniston unassisted, Coniston assisted, Daisy Lake unassisted, and Daisy Lake assisted.

unassisted sites in the same relative position from the historic smelter.

Sampling Procedure

Insects were collected using ramp pitfall traps following methods based on Bouchard et al. (2000) and Pearce et al. (2005). Traps consisted of 473 mL covered food containers with two side holes for entry of ramp and insects. Anti-slip granulated paint was used on the surface of the ramps and a combination of propylene glycol (RV plumbing antifreeze), water, and soap acted as the trap mixture. Three stations were marked at each site and distributed at 25-m intervals along a linear transect to avoid sampling bias (Ward et al. 2001). Each station was associated with four pitfall traps. This resulted in 12 traps per site along a 75-m transect perpendicular to the slope of the hill.

Trap samples were collected every week for 5 weeks in June and July of 2006. The ant community and higher insect taxa were sampled for 1 week in 2007, whereas the beetle community was further sampled for 5 weeks in 2007 to increase its sample size. Plant community data were collected in 2006 using 1×1 -m quadrats along a 100-m transect parallel to the pitfall trap transect, except the Coniston assisted site that contained a 50-m transect. Identification of plant species within each quadrat was determined on-site. Preliminary results from data collected on these sites over the past 4 years (M. Anand,

unpublished data) suggest that the plant community does not change significantly in time on a yearly basis.

Sorting, Preserving, and Identifying

Only adults were selected for further study, whereas adult male ants and queen ants were also excluded for genera analyses, as they do not necessarily represent the presence of a colony and most taxonomic research is based on the workers (Longino et al. 2002). Specimens were identified to various taxonomic levels and were either housed in 75% ethanol or pinned. We identified all specimens to family or morphofamily using methods in Borror et al. (1989). Ants and carabids were then sorted and sent to specialists for further identification. Ants were identified to genus whereas carabids were identified to species.

Data Analysis

Ants often present a problem for quantitative studies, as their social behavior promotes a clumping of individuals within samples. Two common methods to address this issue are to use incidence (presence/absence) data (Longino et al. 2002) or to transform abundance to an ordinal scale (Hoffmann et al. 2000). For this study, we use incidence data for the analyses comparing ant communities with those of plants. Higher insect taxa and carabid species were examined by their abundance.

Ants were removed from the higher taxa analyses due to their ubiquitous presence and high abundance.

Estimated richness was calculated using EstimateS version 7.5.2 for Mac (Colwell 2005). Each sample (trap or quadrat) was iterated 50 times to obtain SDs for Coleman rarefaction (Colwell & Coddington 1994; Colwell et al. 2004). Although past studies using incidence data of ants have had successful results using incidence-based coverage estimate (ICE) compared to other richness estimators (Longino et al. 2002), rarefaction allows for a comparison of species accumulation curves between sites of unequal sampling efforts (Gotelli & Colwell 2001). Shannon diversity and richness values were determined for the samples within each site and compared by Wilcoxon signed-rank test. Patterns in community structure (relative abundance and taxa identity) were further examined for select taxa through nonmetric multidimensional scaling (NMDS) using Statistica 6 (Statsoft Inc. 2001). For NMDS analyses, insect data were pooled into stations for each year resulting in six stations for each site. For the vegetation NMDS analyses, quadrats were grouped every 10 m along the 100-m transect to provide 10 data points for each site. The Coniston assisted site only had 50-m transects, resulting in fewer data points.

Results

A total of 7,912 individuals of 115 morphotaxa were examined for the higher taxa analyses, excluding the ants (Formicidae) (Table 1). For the lower taxa analyses, there were 12 genera of ants that were analyzed and 23 carabid species (Table 2). The taxa found exclusively on assisted sites tended to be phytophages. There were at least seven families of Hemiptera/Homoptera, two Orthoptera, one Thysanoptera, one Dermaptera, and two Coleoptera that are deemed herbivores (Borror et al. 1989) at these restored sites. The ant genus *Temnothorax* is often associated with aphid honeydew (requiring a plant component) (Coovert 2005) and all members of carabid tribe Harpalini (*Bradycellus* and *Harpalus* sp.) are considered phytophages (Lovei & Sunderland 1996).

The Shannon diversity index of the ant community was significantly higher in the Coniston assisted site than in the Coniston unassisted site ($p < 0.001$, $Z = -5.7$). However, there were no significant differences found in the Daisy Lake sites (Fig. 2a & 2b). The Shannon diversity of the carabid community revealed no significant changes (Fig. 2c & 2d). In comparison, there was a significant increase in the plant species at both the Coniston ($p < 0.01$, $Z = 2.94$) and Daisy Lake sites ($p < 0.01$, $Z = 3.62$) (Fig. 3a & 3b). This trend was also observed at the family level for the plants at both the Coniston ($p < 0.01$, $Z = 2.94$) and Daisy Lake sites ($p < 0.05$, $Z = 2.51$) (Fig. 3c & 3d).

The Coleman rarefaction suggests that our sampling effort was sufficient when examining ant genera (Fig. 4a & 4b). Once again, the Coniston assisted site exhibited a higher estimated richness than the unassisted site. The Daisy Lake sites, however, did not show a significant change in estimated

Table 1. Occurrence of insect morphotaxa within the four sites.

	C	CA	DL	DLA
Blattodea	x	x	x	x
Coleoptera A*		x		
Coleoptera B*			x	
Coleoptera C	x	x	x	
Coleoptera D	x	x	x	x
Coleoptera: Anthicidae		x		
Coleoptera: Byrrhidae	x	x	x	x
Coleoptera: Cantharidae*				x
Coleoptera: Carabidae	x	x	x	x
Coleoptera: Cerambycidae*			x	
Coleoptera: Chrysomelidae	x	x	x	x
Coleoptera: Ciidae*	x			
Coleoptera: Cleridae	x			x
Coleoptera: Coccinellidae	x	x		x
Coleoptera: Cucujidae*			x	
Coleoptera: Curculionidae	x	x	x	x
Coleoptera: Elateridae	x	x	x	x
Coleoptera: Eucinetidae		x		
Coleoptera: Hydrophilidae		x		x
Coleoptera: Lathridiidae	x	x	x	x
Coleoptera: Leiodidae	x	x		x
Coleoptera: Melandryidae			x	x
Coleoptera: Mordellidae	x	x	x	x
Coleoptera: Nitidulidae		x	x	x
Coleoptera: Pselaphidae	x	x	x	x
Coleoptera: Ptilidae*				x
Coleoptera: Scarabidae	x	x	x	x
Coleoptera: Scolytidae*		x		
Coleoptera: Scydmaenidae	x	x	x	x
Coleoptera: Staphylinidae	x	x	x	x
Coleoptera: Tenebrionidae*		x		
Coleoptera: Throscidae	x	x		
Coleoptera: unknown	x		x	
Dermaptera		x		
Diptera A		x	x	
Diptera B		x		x
Diptera C*				x
Diptera D*		x		
Diptera: Anisopodidae	x		x	x
Diptera: Anthomyiidae	x	x	x	x
Diptera: Asilidae*	x			
Diptera: Ceratopogonidae	x	x	x	x
Diptera: Chloropidae	x	x	x	x
Diptera: Culicidae	x	x	x	x
Diptera: Dolichopodidae		x	x	x
Diptera: Drosophilidae	x	x	x	x
Diptera: Empididae	x			
Diptera: Heleomyzidae	x	x	x	x
Diptera: Lauxaniidae	x			x
Diptera: Muscidae	x	x	x	x
Diptera: Mycetophilidae	x	x	x	x
Diptera: Nematocera	x	x	x	x
Diptera: Phoridae	x	x	x	x
Diptera: Psychodidae	x		x	
Diptera: Sarcophagidae	x	x	x	x
Diptera: Scathophagidae*			x	
Diptera: Sphaeroceridae	x	x	x	x

Table 1. Continued

	C	CA	DL	DLA
Diptera: Syrphidae*	x			
Diptera: Tachinidae		x	x	x
Diptera: Tipulidae*	x			
Diptera: Unknown	x		x	x
Hemiptera: Aradidae*		x		
Hemiptera: Coreidae	x	x		
Hemiptera: Lygaeidae	x	x	x	x
Hemiptera: Miridae		x	x	x
Hemiptera: Nabidae	x	x	x	x
Hemiptera: Pentatomidae		x	x	
Hemiptera: Rhopalidae*		x		
Hemiptera: Tingidae	x	x	x	x
Hemiptera: Unknown		x		x
Homoptera: Aphididae	x	x	x	x
Homoptera: Cercopidae		x	x	x
Homoptera: Cicadellidae	x	x	x	x
Homoptera: Cixiidae	x	x	x	x
Homoptera: Coccidae		x		x
Homoptera: Delphacidae*				x
Homoptera: Dictyopharidae*		x		
Homoptera: Eriosomatidae	x	x	x	x
Homoptera: Issidae		x		x
Homoptera: unknown		x		x
Hymenoptera A	x	x		
Hymenoptera B	x			
Hymenoptera C*	x			
Hymenoptera D*		x		
Hymenoptera E	x	x		
Hymenoptera: Apidae	x	x		
Hymenoptera: Braconidae	x	x	x	x
Hymenoptera: Chalcidoidea	x	x	x	x
Hymenoptera: Colletidae	x	x		
Hymenoptera: Diapriidae	x	x	x	x
Hymenoptera: Halictidae	x		x	x
Hymenoptera: Ichneumonidae	x	x	x	x
Hymenoptera: Megachilidae	x	x		x
Hymenoptera: Mutillidae*			x	
Hymenoptera: Myrmariidae*	x			
Hymenoptera: Pompilidae	x	x	x	x
Hymenoptera: Scelionid (Idris)	x	x	x	x
Hymenoptera: Sphecidae		x	x	
Hymenoptera: unknown	x	x	x	x
Hymenoptera: Vespidae	x	x	x	x
Lepidoptera	x	x	x	x
Neuroptera*				x
Odonata*	x			
Orthoptera A		x		x
Orthoptera B	x	x	x	x
Orthoptera: Acrididae		x		
Orthoptera: Gryllacrididae	x	x		
Orthoptera: Gryllidae	x	x	x	x
Orthoptera: Tetrigidae	x	x		x
Orthoptera: unknown	x			x
Psocoptera	x	x	x	x
Thysanoptera: Aeolothripidae*		x		
Thysanoptera: Phlaeothripidae	x	x	x	x
Thysanoptera: Thripidae	x	x	x	x
Trichoptera	x			
Total richness	77	88	66	74

C, Coniston; CA, Coniston assisted; DL, Daisy Lake; DLA, Daisy Lake assisted.

* Denotes taxa represented by one specimen.

Table 2. Occurrence of ant and carabid taxa within the four sites.

Taxa	C	CA	DL	DLA
Ants (Formicidae)				
<i>Formica</i>	X	X	X	X
<i>Myrmica</i>	X	X	X	X
<i>Lasius</i>	X		X	X
<i>Ponera</i>		X		X
<i>Aphaenogaster</i>			X	
<i>Acanthomyops</i> *			X	
<i>Camponotus</i>	X	X	X	X
<i>Crematogaster</i>	X	X	X	X
<i>Dolichoderus</i>		X	X	X
<i>Temnothorax</i>		X		X
<i>Brachymyrmex</i>		X	X	X
<i>Tapinoma</i>	X	X	X	X
Total ant richness	6	9	10	10
Carabids (Carabidae)				
<i>Agonum cupripenne</i>			X	
<i>Amara angustata</i>		X	X	X
<i>Amara obesa</i> *		X		
<i>Apristus subsulcatus</i>	X	X		X
<i>Bradycellus nigrinus</i> *		X		
<i>Calathus ingratus</i>		X		
<i>Calosoma calidum</i>	X	X		
<i>Calosoma frigidum</i>				X
<i>Cicindela longilabris</i>			X	
<i>Cicindela sexguttata</i>	X		X	X
<i>Cymindis cribricollis</i>	X	X		X
<i>Cymindis neglecta</i>	X	X	X	
<i>Diplocheila obtuse</i>		X		
<i>Dyschirius sphaericollis</i> *			X	
<i>Harpalus opacipennis</i>		X		
<i>Harpalus somnulentus</i>		X		X
<i>Lebia moesta</i> *				X
<i>Lebia pumila</i> *			X	
<i>Microlestes linearis</i> *		X		
<i>Miscodera arctica</i>	X			
<i>Poecilus lucublandus</i>		X		X
<i>Pterostichus mutus</i>	X	X	X	X
<i>Synuchus impunctatus</i>	X		X	X
Total carabid richness	8	14	9	10

C, Coniston; CA, Coniston assisted; DL, Daisy Lake; DLA, Daisy Lake assisted.

* Denotes taxa represented by one specimen.

richness between assisted and unassisted sites. The sampling effort of the carabids appears to be not as robust, as the rarefaction does not reach an asymptote. Nevertheless, the rarefaction graph shows the estimated carabid richness of both assisted sites would to be larger than the unassisted sites (Fig. 4c & 4d).

In terms of species richness, the plant species show a significant increase in assisted sites for both the Coniston sites ($p < 0.01$, $Z = 10.11$) and the Daisy Lake sites ($p < 0.01$, $Z = 7.42$) (Fig. 5a & 5b). Furthermore, the Coniston unassisted site exhibited significantly lower plant richness than the Daisy Lake unassisted sites ($p < 0.05$, $Z = 2.5$) whereas the Coniston assisted site exhibited significantly higher plant richness than the Daisy Lake assisted site ($p < 0.01$, $Z = 8.0$).

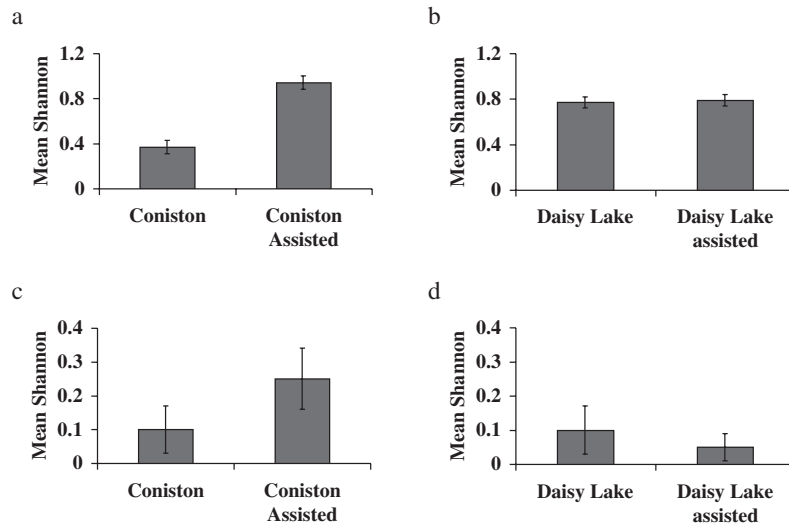


Figure 2. Mean Shannon diversity values with SE of the mean of ant genera (a,b) and carabid beetle species (c,d). Significant at * $p < 0.05$; ** $p < 0.01$.

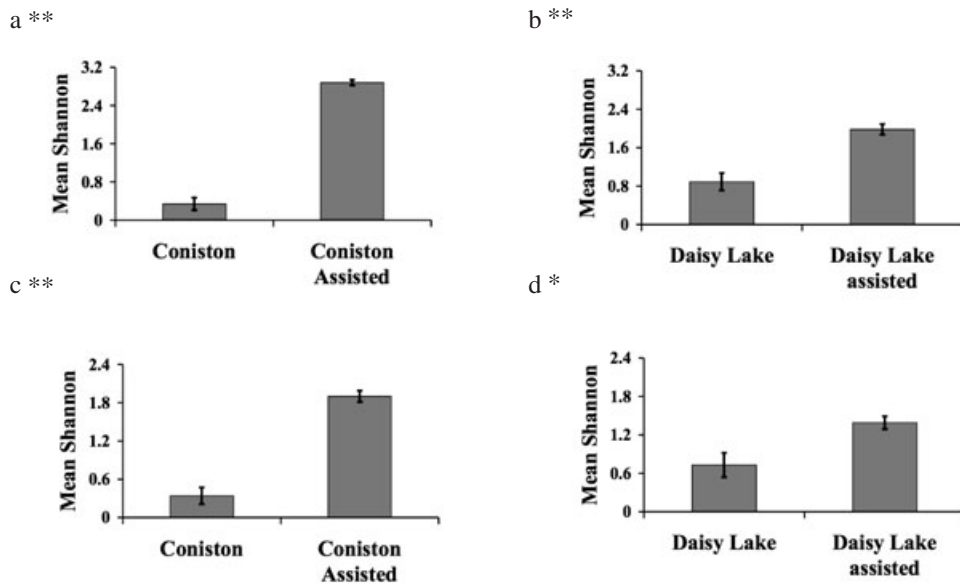


Figure 3. Mean Shannon diversity values with SE of the mean of plant species (a,b) and plant families (c,d). Significant at * $p < 0.05$; ** $p < 0.01$.

The insect morphofamilies exhibited a significant increase in richness in the Coniston assisted site ($p < 0.05$, $Z = -2.20$) but no difference was observed in the Daisy Lake site ($p > 0.05$) (Fig. 5c & 5d). The ant genera exhibited a significant increase in richness in the Coniston assisted site ($p < 0.01$, $Z = -5.84$) but no difference in the Daisy Lake site ($p > 0.05$) (Fig. 5e & 5f). The carabid species exhibited no significant difference between any of the sites ($p > 0.05$) (Fig. 5g & 5h).

When examining the ant community structure data using NMDS, there was distinct clustering of stations at the Coniston sites but not the Daisy Lake sites, revealing a separation between the Coniston assisted site and the other sites

(Fig. 6a–c). The carabid data, once again, did not reveal a distinguishing pattern between sites for either Coniston or Daisy Lake. Three stations were considered outliers for the NMDS of the carabids because their presence created a graphical barrier in examining the patterns for the other stations. That is to say, they were so distinct in three different directions that the rest of the stations clustered tightly together. The presence of these outliers was due to combining the traps (used for other analyses) of the stations, resulting in three stations with a larger number of beetles than the others; one from Daisy Lake unassisted and two from Daisy Lake assisted. By removing these three outliers, the pattern remained the same in that there were no differences in the composition between

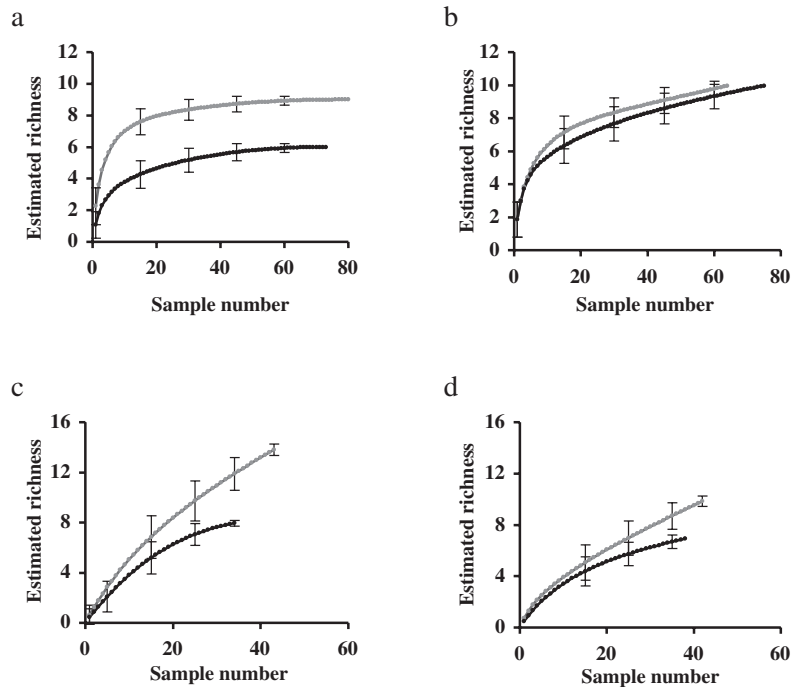


Figure 4. Coleman rarefaction of the ant community at the Coniston sites (a), ants at the Daisy Lake sites (b), carabid beetle community at the Coniston sites (c), and carabid beetles at the Daisy Lake sites (d). Black lines represent unassisted sites and assisted sites are gray. Error bars represent one SD.

the sites (Fig. 6d–f). The NMDS for the vegetation reveals that there are community differences between the two Coniston sites, between the two Daisy Lake sites and between the Coniston assisted site and all other sites (Fig. 6g–i).

Discussion

We found that although plant (species and family) diversity increased significantly in assisted areas compared to unassisted ones, insect (various taxa) diversity did not. Restoration efforts were thus only moderately successful for reestablishing terrestrial insect communities at the Coniston site compared to the recovery of the plant community. The Daisy Lake site did not reveal successful restoration for the terrestrial insects in any of the examined taxa. Thus, we conclude that restoration of plant communities, in terms of plant diversity, may not be sufficient for bringing about ecosystem-level changes, including the restoration of insect communities. Although our results suggest that the Daisy Lake sites are already restored in terms of the insect community, the NMDS of the ant and vegetation communities clearly shows that the composition of the Daisy Lake sites and the Coniston unassisted site is similar, whereas, the Coniston assisted site is quite different. This shows that the restoration efforts at the Coniston site are indeed changing the ant and vegetation community composition at this site. This trend may also be observed by examining the taxa of the sites. There were more taxa unique to the Coniston site and included two saprovores: Dermaptera (earwigs) and Eucinetidae (plate-thigh beetles). Only two

taxa were represented by more than one individual and were uniquely found at the Daisy Lake assisted site and both are primarily considered predators: Cleridae (Checkered beetles) and *Calosoma frigidum* (Caterpillar hunter) (Borror et al. 1989). This may suggest more availability of leaf litter (humus layer) at the Coniston site than the Daisy Lake site to support saprovores. We therefore find that the Coniston assisted site has been moderately successful in reestablishing some insect taxa but the Daisy Lake site has not.

The differences between the technique and timing of restoration of the two assisted sites may account for the differences observed between the sites. Future studies may reveal whether aerial liming is less effective than the manual liming or if an extra decade of restoration would result in significant changes. Indeed, studies show that the rehabilitation of insect species assemblages from a disturbed state to the expected original state may only take several years; however, the trend is not always linear or observed across different taxa due to successional changes in the habitat (Davis et al. 2002, 2003, Wassenaar et al. 2005). The discrepancies between Coniston and Daisy Lake, however, do not explain why there is only a marginal improvement in certain aspects of the insect community for either site. We therefore suggest reasons why either of the restoration techniques may not be successful for the insect community.

Although restoration efforts provide suitable habitat and food for certain herbivorous insects, they may not enhance forest structural complexity, which includes attributes such as foliage arrangement, tree height, and volume of large woody debris (McElhinny et al. 2005). The structural heterogeneity

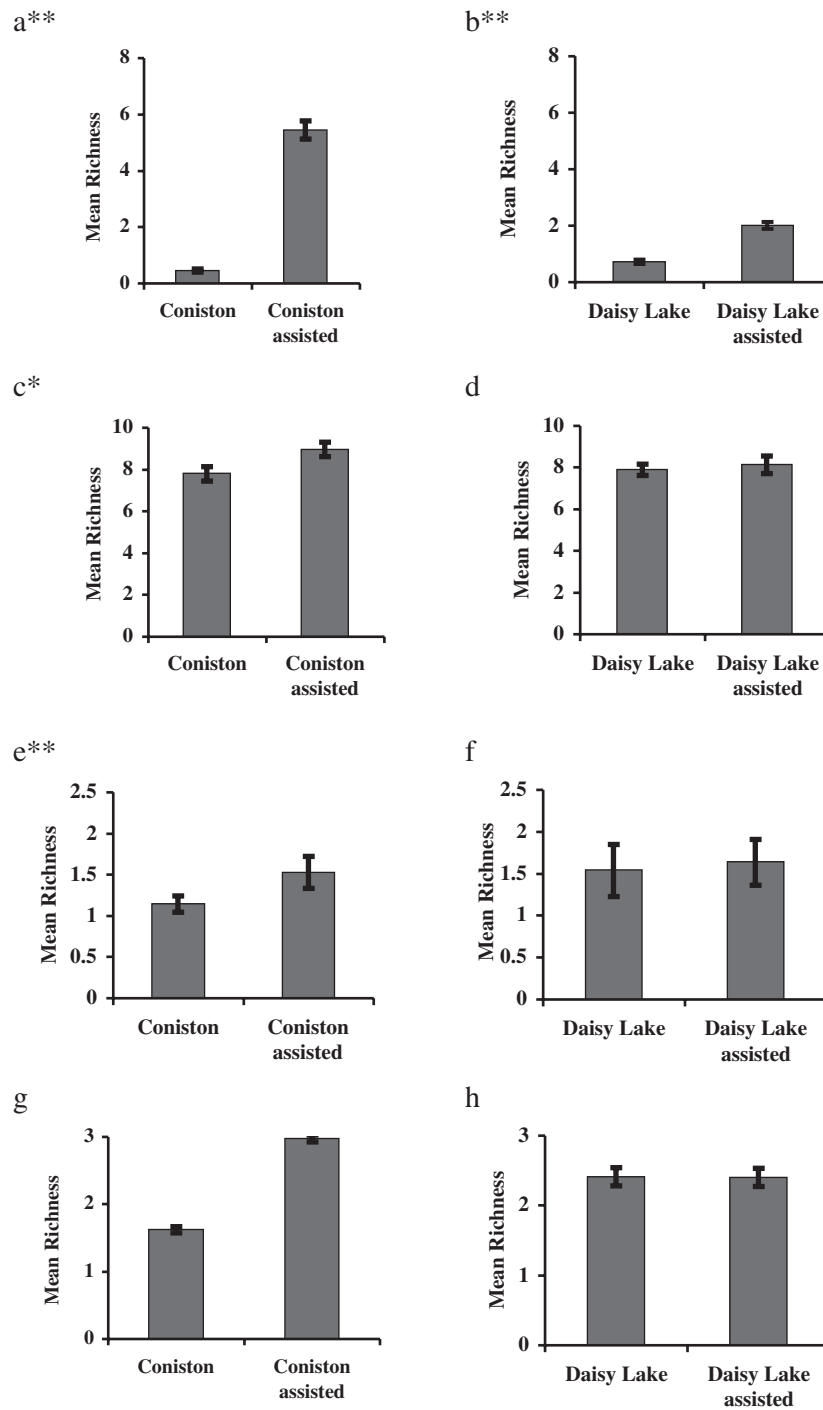


Figure 5. Mean richness for plants (a,b), all insect morphotaxa (c,d), ants (e,f), and carabids (g,h) with SE of the mean. Significant at * $p < 0.05$; ** $p < 0.01$.

hypothesis specifies that the structural complexity of the plant community may be more important in determining the richness of the insects (or other higher trophic groups) than the number of plant species (Brose 2003; Harvey et al. 2008). Industrially disturbed sites, such as the ones studied here, are often devoid of any woody debris or even a humus layer preferred by,

or required for, many scavengers, detritivores, predaceous species, or other feeding guilds. Such aspects of the structural complexity are not easily reestablished with the traditional restoration techniques used for these sites.

The presence of herbivorous carabids (*Bradycellus* and *Harpalus* sp.) at both assisted sites may also affect the

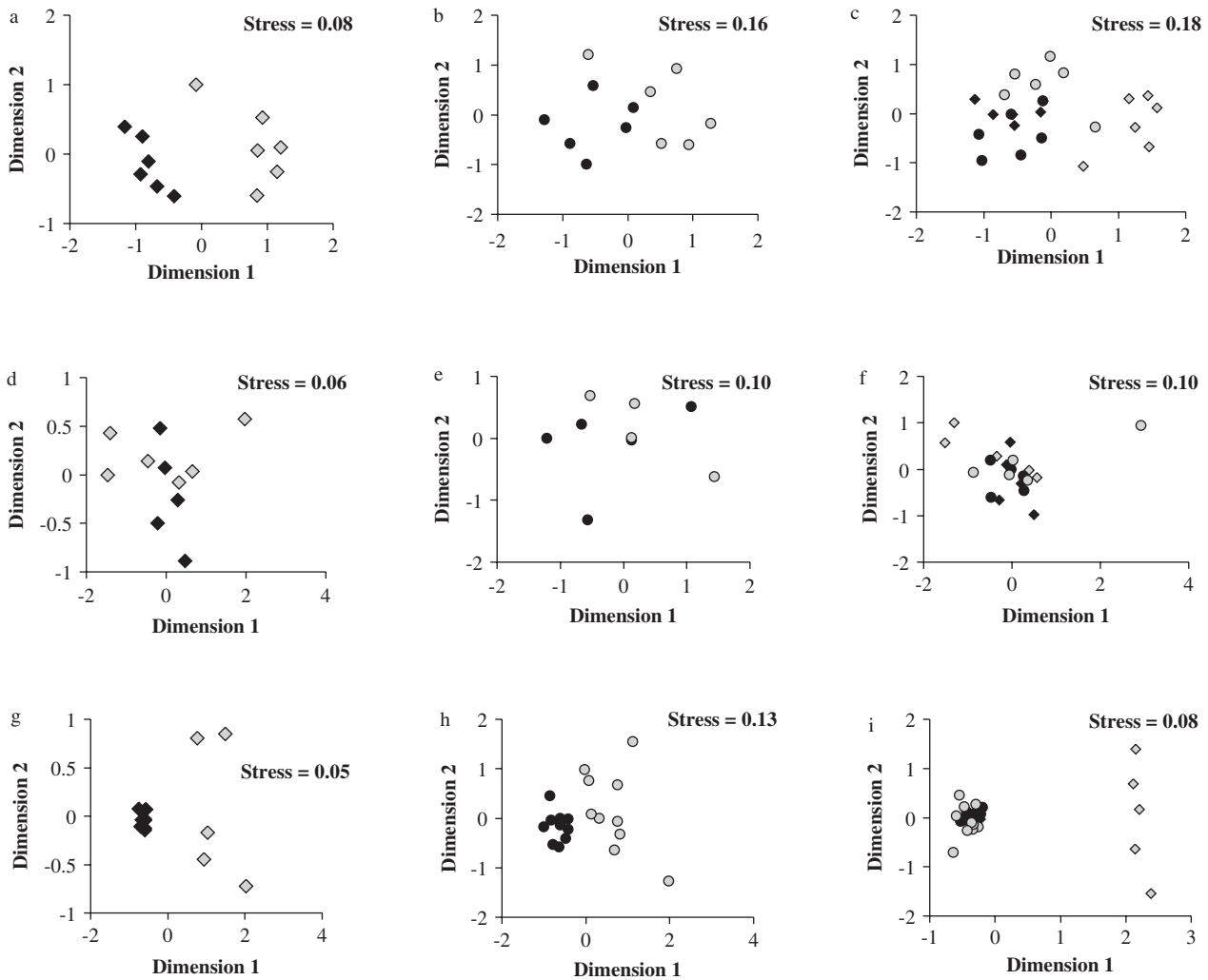


Figure 6. NMDS plots showing differences between sampling stations (all sites) based on ant genera incidence (a–c), carabid species abundance (d–f) and plant species (g–i). Diamonds represent Coniston sites whereas circles represent Daisy Lake sites. Black symbols represent unassisted sites and gray symbols represent assisted sites.

success of restoration efforts. Studies have shown that other seed-eating carabid species are effective granivores that may limit herbaceous weed populations (Honek et al. 2009). Such granivores may hinder the rehabilitation process or perhaps aid it by removing some non-native species preferentially. Another reason for the lack of response of the insect community could simply be the fact that we could not capture the entire spectrum of insects. Our study focused on insects that are found in severely disturbed landscapes and that are susceptible to pitfall trapping. The community captured by pitfall traps is an important one (Pearce et al. 2005) and thus we feel it provides a reasonable baseline study, the first, to our best knowledge, to examine restoration effects on insects in this area.

We found that plant diversity, both at the species and family levels, increased in both assisted sites whereas certain insect taxa diversity increased at the Coniston site but not for the

Daisy Lake site. We therefore cannot support the idea that plant diversity may be used as surrogate for insect diversity in this region. Within the insect community itself, cross-taxon surrogacy is not always appropriate or reliable because insects of different guilds have such a diversity of environmental and nutritional requirements (Negi & Gadgil 2002; Lovell et al. 2007). A multi-taxa approach, however, may provide different surrogates for groups of insects that share habitat, feeding strategies, or guilds (Lovell et al. 2007). Using higher taxa as surrogates for the species level has been more successful by having stronger congruency between the groups (Negi & Gadgil 2002; Lovell et al. 2007; Cardoso et al. 2008). Therefore, rapid biodiversity assessments performed by using higher taxa may still provide useful information for conservation and biodiversity studies.

In this study, we showed that both the insect family level and ant genus level counts revealed a marginal improvement

and change in composition from the restoration efforts at the Coniston site but not at the Daisy Lake sites. There were no significant differences for the carabid species (primarily predators). We suggest that if there are large changes in the community, such as those found at the Coniston sites, higher taxa surrogates would be sufficient to detect them. In addition, the restoration efforts may not be as effective for predators, such as most carabids, compared to generalists like ants (Spalding & Haes 1995). The differences between the family and genera analyses and the carabid species analyses show how important examining different guilds would be in such a study. Insect families and ant genera exhibited similar results, showing larger diversity changes in the Coniston sites than in the Daisy Lake sites. Thus, we conclude that these insect taxa can be used as surrogates for other insects, increasing the feasibility of future entomology studies in the area.

The lack of significant results for the carabid species is also an important component of this study as it examines the effects of restoration on predatory insects. Restoration techniques change the structure of the habitat by adding a herbaceous layer to barren rock that makes it suboptimal for visual predators such as many carabids (Spalding & Haes 1995; Harvey et al. 2008). In fact, there are arguments that degraded landscapes of industrial perturbation may be considered unique habitats that should be protected: benefits of the barren land include sun exposure for basking insects like butterflies and grasshoppers and space for fast running predators (Spalding & Haes 1995; Kozlov & Zvereva 2007). Therefore, it is not surprising that a generalist group like ants would benefit from restoration efforts whereas the carabids do not.

Restoration techniques in this area could be further adapted to help to accelerate the recovery of both plants and insects. Recently, it has been discovered that physical sheltering by wooden fences or natural rock can increase the survival and performance of some tree species (in this case, mountain birch seedlings in Monchegorsk, Russia) (Eränen & Kozlov 2006). Another important area of research relates to enhancing the humus layer to increase the natural recolonization of pioneer species and increase the survival of planted seedlings/shrubs, which has been examined in Harjavalta, Finland by adding mulch to the soil (Helmisaari et al. 2007). These and other methods of restoring forest floor complexity (by adding woody debris) can provide areas for fauna to escape predators and the elements and may help to accelerate insect diversity. Majer (1989) gives a list of various wildlife rehabilitation techniques that may provide habitat and shelter for fauna of reclaimed land. These techniques include adding direct-haul topsoil, creating large boulder and rock piles, creating large brush piles, and implanting dead trees or large woody debris. Only a handful of mining companies have taken the invertebrate fauna into consideration in restoration efforts (Majer et al. 2007), but we suggest that it should become of increasing importance.

We realize that the ultimate goal of restoration is not always simply to increase species diversity. Other factors must be taken into consideration such as ecosystem functioning (e.g. nutrient cycling, soil formation) and ecosystem services (e.g. pollination, wood production) (Ehrenfeld 2000; Ruiz-Jaen

& Aide 2005). Although there is a relationship between diversity and ecosystem function, the strength of the relationship is often under debate (Srivastava & Vellend 2005). Further studies could examine more explicitly the roles of both plant and insect communities in ecosystem function in the context of restoration.

Implications for Practice

- Insect diversity and richness do not mirror the plant diversity or richness and therefore, the use of plant diversity or richness for restoration success should be cautionary in landscapes devastated by copper–nickel smelter operations and perhaps other perturbed areas.
- Terrestrial arthropod fauna should be incorporated as an important indicator of restoration success for landscapes devastated by copper–nickel smelter operations and perhaps other perturbed areas.
- Rapid assessment using the most abundant insect families may be sufficient for certain aspects of restoration in such landscapes.
- More detailed analyses may use ants (Hymenoptera: Formicidae) and carabid beetles (Coleoptera: Carabidae) as useful and feasible insect taxa as surrogates for generalist and predatory insect guilds, respectively.

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